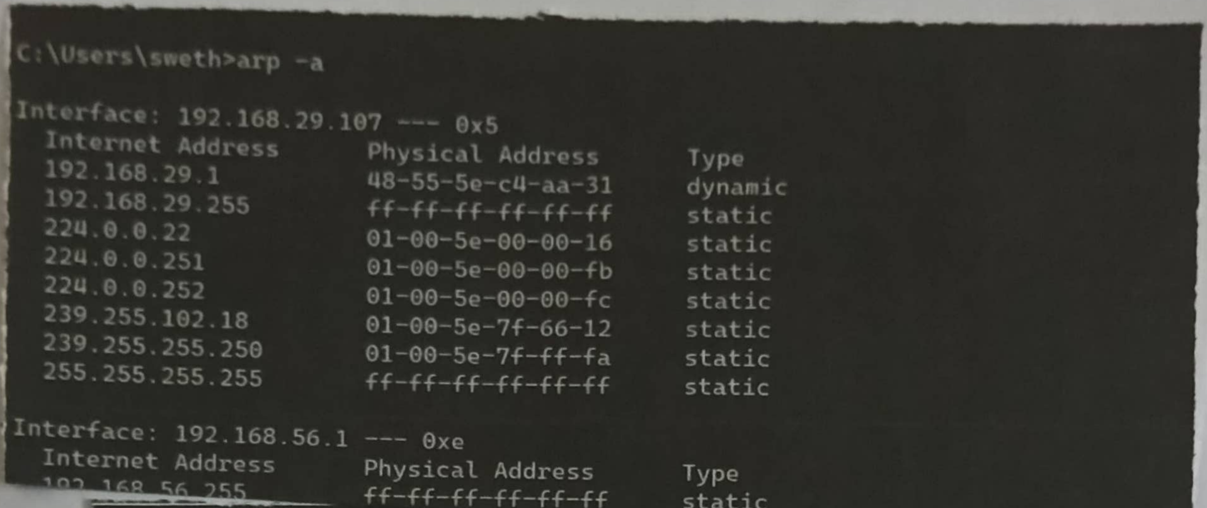
16/7/26

Exp-1 Basic Networking Commands.

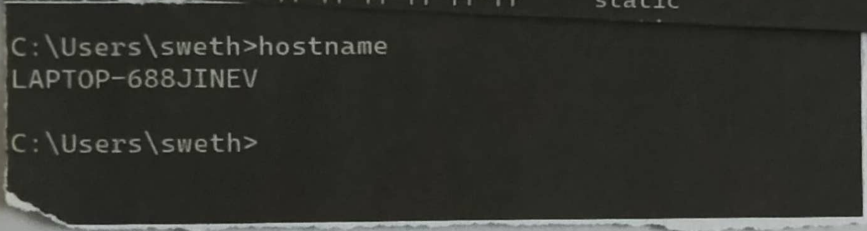
WINDOWS NETWORKING COMMANDS

1. arp - a - It will show the IP address of your computer



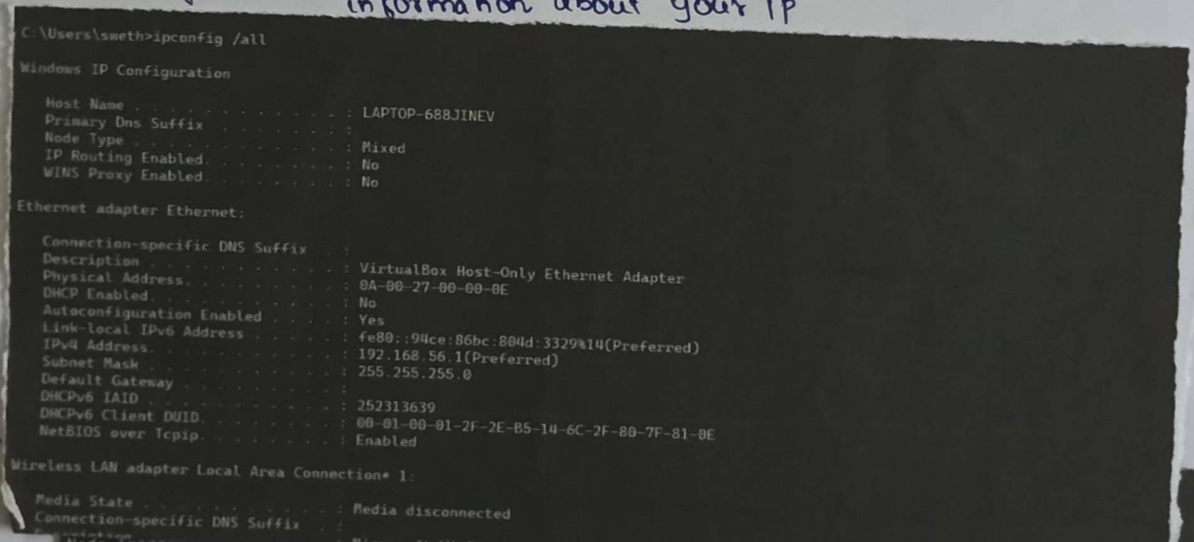
```
C:\Users\sweth>arp -a

Interface: 192.168.29.107 --- 0x5
  Internet Address      Physical Address      Type
  192.168.29.1          48-55-5e-c4-aa-31     dynamic
  192.168.29.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.56.1 --- 0xe
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
```

2.



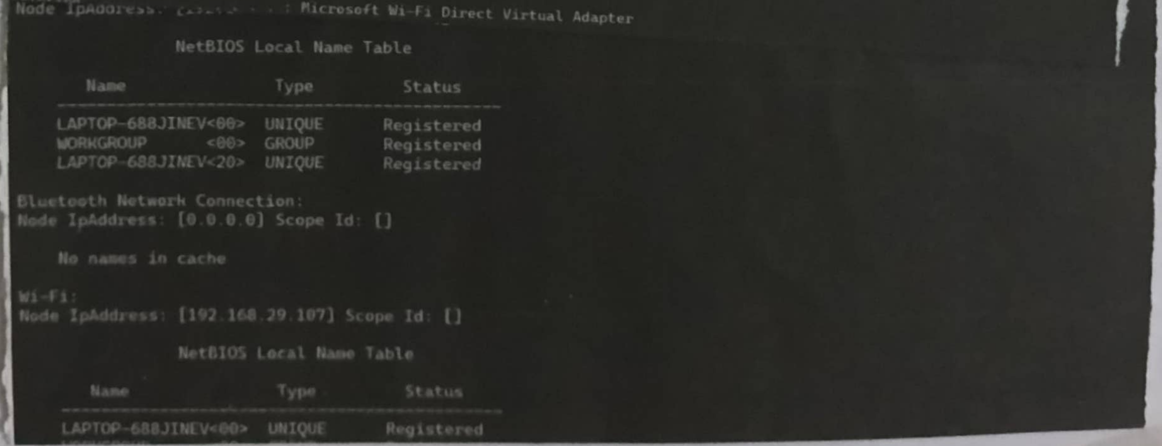
```
C:\Users\sweth>hostname
LAPTOP-688JINEV

C:\Users\sweth>
```

3. ipconfig /all - This command displays detailed configuration information about your IP



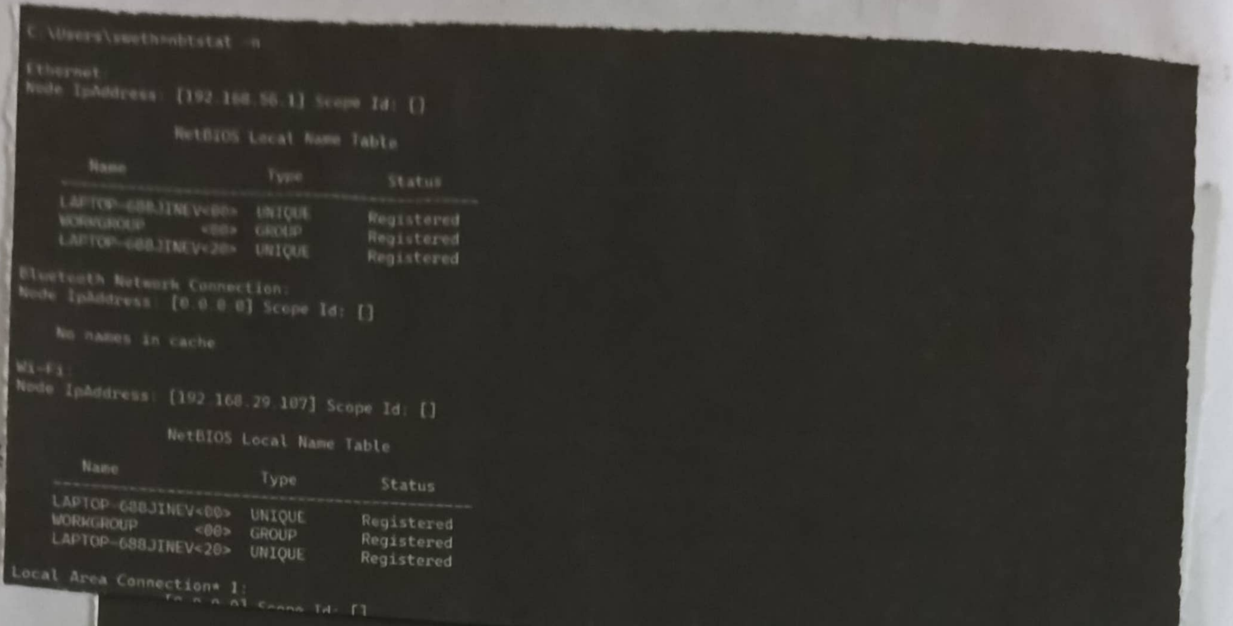
```
C:\Users\sweth>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : LAPTOP-688JINEV
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Mixed
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-0E
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::94ce:86bc:804d:3329%14(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 252313639
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2F-2E-B5-14-6C-2F-80-7F-81-0E
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

4.

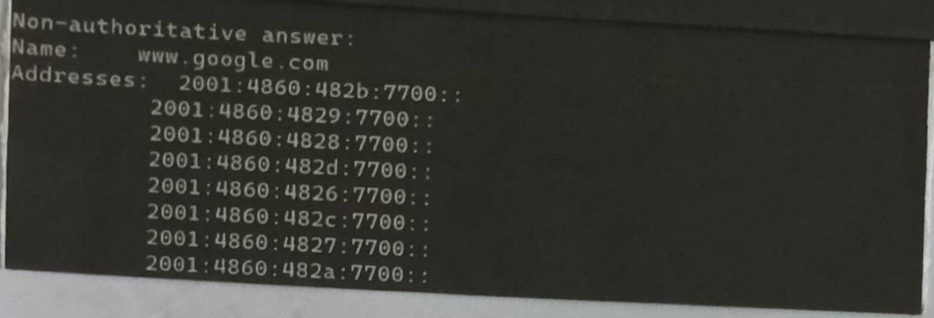


```
Node IpAddress. . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter

                NetBIOS Local Name Table

       Name               Type         Status
    ---------------------------------------------
    LAPTOP-688JINEV<00>  UNIQUE      Registered
    WORKGROUP      <00>  GROUP       Registered
    LAPTOP-688JINEV<20>  UNIQUE      Registered

Bluetooth Network Connection:
Node IpAddress: [0.0.0.0] Scope Id: []

    No names in cache

Wi-Fi:
Node IpAddress: [192.168.29.107] Scope Id: []

                NetBIOS Local Name Table

       Name               Type         Status
    ---------------------------------------------
    LAPTOP-688JINEV<00>  UNIQUE      Registered
```

5. nbtstat - a

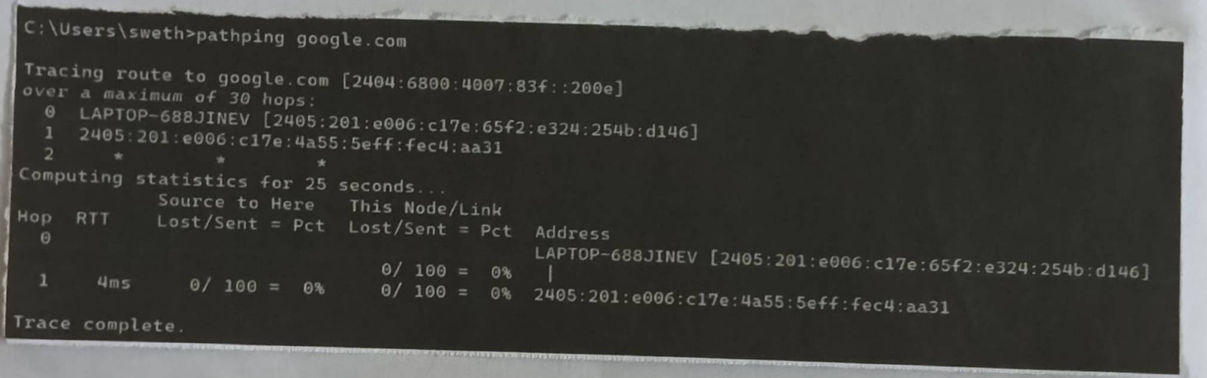


```
C:\Users\sweth>nbtstat -n

Ethernet:
Node IpAddress: [192.168.56.1] Scope Id: []

                NetBIOS Local Name Table

       Name               Type         Status
    ---------------------------------------------
    LAPTOP-688JINEV<00>  UNIQUE      Registered
    WORKGROUP      <00>  GROUP       Registered
    LAPTOP-688JINEV<20>  UNIQUE      Registered

Bluetooth Network Connection:
Node IpAddress: [0.0.0.0] Scope Id: []

    No names in cache

Wi-Fi:
Node IpAddress: [192.168.29.107] Scope Id: []

                NetBIOS Local Name Table

       Name               Type         Status
    ---------------------------------------------
    LAPTOP-688JINEV<00>  UNIQUE      Registered
    WORKGROUP      <00>  GROUP       Registered
    LAPTOP-688JINEV<20>  UNIQUE      Registered

Local Area Connection* 1:
                [0.0.0.0] Scope Id: []
```



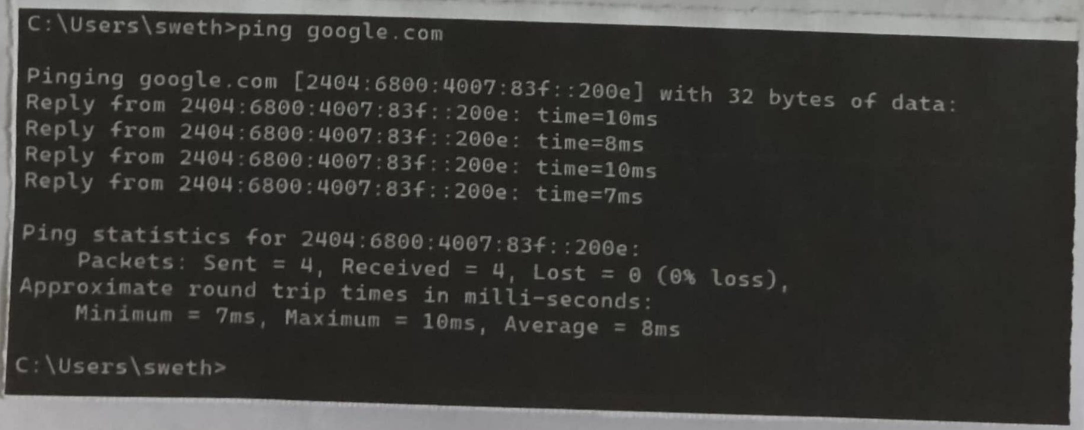
```
Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:482b:7700::
          2001:4860:4829:7700::
          2001:4860:4828:7700::
          2001:4860:482d:7700::
          2001:4860:4826:7700::
          2001:4860:482c:7700::
          2001:4860:4827:7700::
          2001:4860:482a:7700::
```

7. pathping



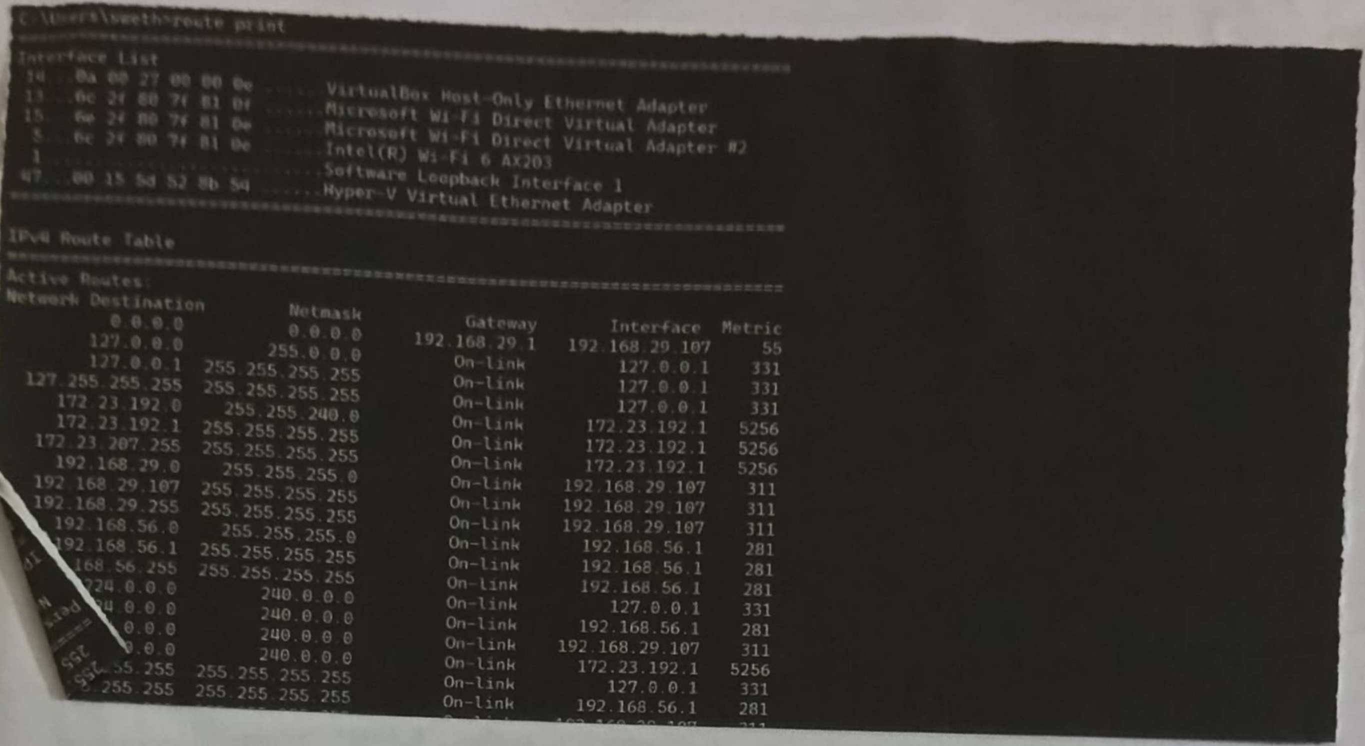
```
C:\Users\sweth>pathping google.com

Tracing route to google.com [2404:6800:4007:83f::200e]
over a maximum of 30 hops:
  0  LAPTOP-688JINEV [2405:201:e006:c17e:65f2:e324:254b:d146]
  1  2405:201:e006:c17e:4a55:5eff:fec4:aa31
  2     *        *        *
Computing statistics for 25 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           LAPTOP-688JINEV [2405:201:e006:c17e:65f2:e324:254b:d146]
                                0/ 100 =  0%   |
  1    4ms     0/ 100 =  0%     0/ 100 =  0%  2405:201:e006:c17e:4a55:5eff:fec4:aa31

Trace complete.
```

8. ping



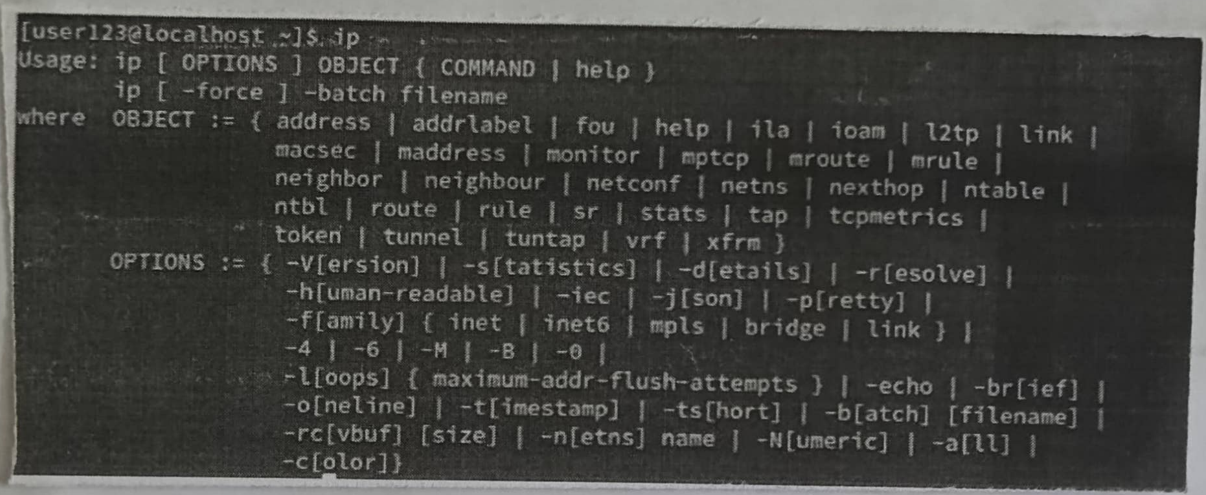
```
C:\Users\sweth>ping google.com

Pinging google.com [2404:6800:4007:83f::200e] with 32 bytes of data:
Reply from 2404:6800:4007:83f::200e: time=10ms
Reply from 2404:6800:4007:83f::200e: time=8ms
Reply from 2404:6800:4007:83f::200e: time=10ms
Reply from 2404:6800:4007:83f::200e: time=7ms

Ping statistics for 2404:6800:4007:83f::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 10ms, Average = 8ms

C:\Users\sweth>
```

9. Route



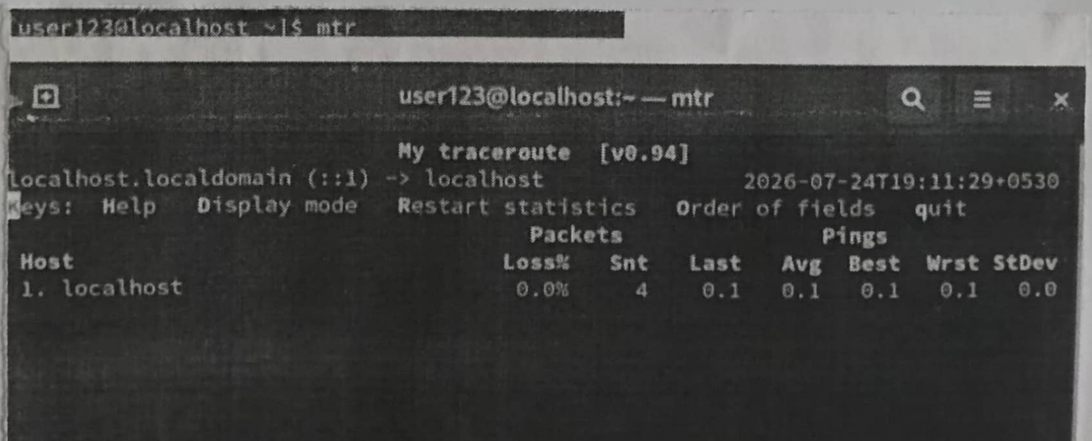
```
C:\Users\sweth>route print
===========================================================================
Interface List
 14...0a 00 27 00 00 0e ......VirtualBox Host-Only Ethernet Adapter
 13...6c 2f 80 7f 81 0f ......Microsoft Wi-Fi Direct Virtual Adapter
 15...6e 2f 80 7f 81 0e ......Microsoft Wi-Fi Direct Virtual Adapter #2
  5...6c 2f 80 7f 81 0e ......Intel(R) Wi-Fi 6 AX203
  1...........................Software Loopback Interface 1
 47...00 15 5d 52 8b 54 ......Hyper-V Virtual Ethernet Adapter
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0     192.168.29.1  192.168.29.107     55
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
     172.23.192.0    255.255.240.0         On-link         127.0.0.1    331
     172.23.192.1  255.255.255.255         On-link      172.23.192.1   5256
   172.23.207.255  255.255.255.255         On-link      172.23.192.1   5256
     192.168.29.0    255.255.255.0         On-link      172.23.192.1   5256
   192.168.29.107  255.255.255.255         On-link    192.168.29.107    311
   192.168.29.255  255.255.255.255         On-link    192.168.29.107    311
     192.168.56.0    255.255.255.0         On-link    192.168.29.107    311
     192.168.56.1  255.255.255.255         On-link      192.168.56.1    281
      168.56.255  255.255.255.255         On-link      192.168.56.1    281
        24.0.0.0        240.0.0.0         On-link      192.168.56.1    281
         4.0.0.0        240.0.0.0         On-link         127.0.0.1    331
           0.0.0        240.0.0.0         On-link      192.168.56.1    281
           0.0.0        240.0.0.0         On-link    192.168.29.107    311
          55.255  255.255.255.255         On-link      172.23.192.1   5256
         255.255  255.255.255.255         On-link         127.0.0.1    331
                  255.255.255.255         On-link      192.168.56.1    281
```

1. ip



```
[user123@localhost ~]$ ip
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where  OBJECT := { address | addrlabel | fou | help | ila | ioam | l2tp | link |
                   macsec | maddress | monitor | mptcp | mroute | mrule |
                   neighbor | neighbour | netconf | netns | nexthop | ntable |
                   ntbl | route | rule | sr | stats | tap | tcpmetrics |
                   token | tunnel | tuntap | vrf | xfrm }
       OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
                    -h[uman-readable] | -iec | -j[son] | -p[retty] |
                    -f[amily] { inet | inet6 | mpls | bridge | link } |
                    -4 | -6 | -M | -B | -0 |
                    -l[oops] { maximum-addr-flush-attempts } | -echo | -br[ief] |
                    -o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename] |
                    -rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
                    -c[olor]}
```

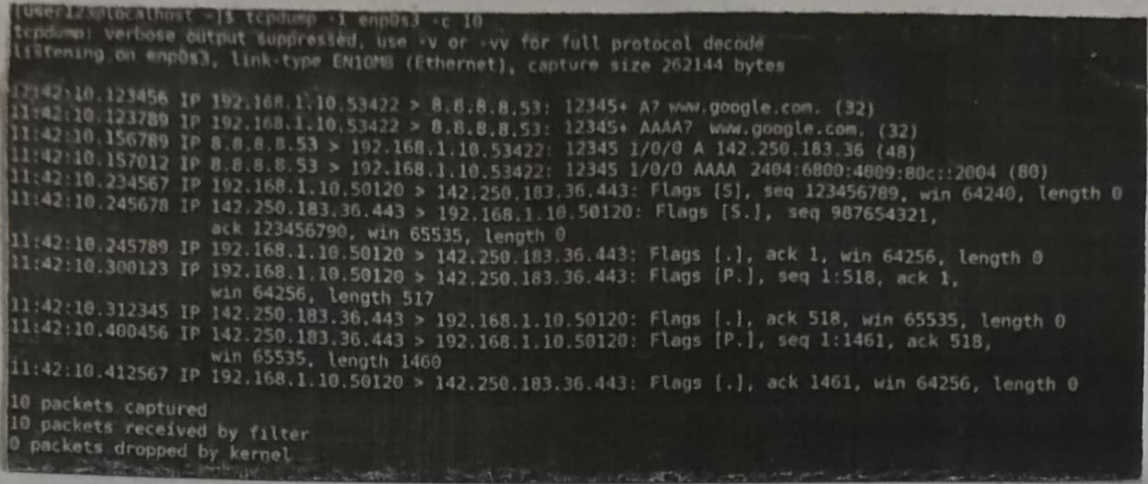
2. mtr



```
user123@localhost ~]$ mtr
```

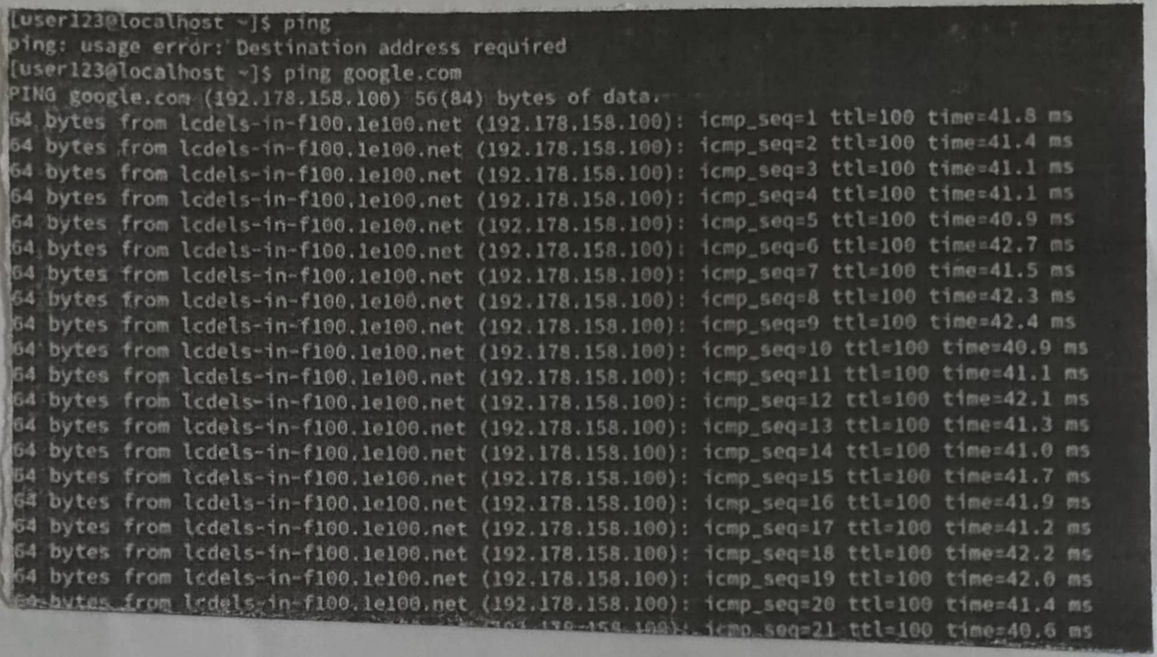
```
                             My traceroute  [v0.94]
localhost.localdomain (::1) -> localhost              2026-07-24T19:11:29+0530
Keys:  Help   Display mode   Restart statistics   Order of fields   quit
                                       Packets               Pings
 Host                                Loss%   Snt   Last   Avg  Best  Wrst StDev
 1. localhost                         0.0%     4    0.1   0.1   0.1   0.1   0.0
```

3. tcpdump



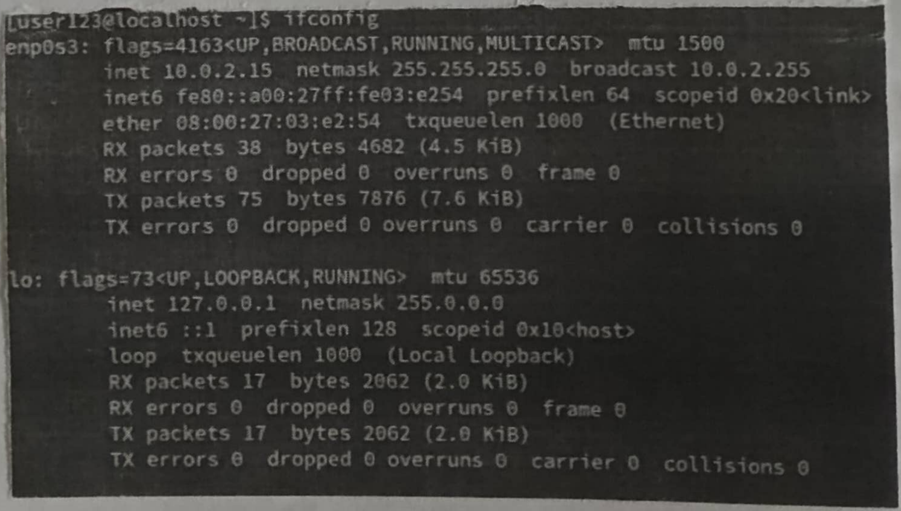
```
[user123@localhost ~]$ tcpdump -i enp0s3 -c 10
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes

11:42:10.123456 IP 192.168.1.10.53422 > 8.8.8.8.53: 12345+ A? www.google.com. (32)
11:42:10.123789 IP 192.168.1.10.53422 > 8.8.8.8.53: 12345+ AAAA? www.google.com. (32)
11:42:10.156789 IP 8.8.8.8.53 > 192.168.1.10.53422: 12345 1/0/0 A 142.250.183.36 (48)
11:42:10.157012 IP 8.8.8.8.53 > 192.168.1.10.53422: 12345 1/0/0 AAAA 2404:6800:4009:80c::2004 (80)
11:42:10.234567 IP 192.168.1.10.50120 > 142.250.183.36.443: Flags [S], seq 123456789, win 64240, length 0
11:42:10.245678 IP 142.250.183.36.443 > 192.168.1.10.50120: Flags [S.], seq 987654321,
                   ack 123456790, win 65535, length 0
11:42:10.245789 IP 192.168.1.10.50120 > 142.250.183.36.443: Flags [.], ack 1, win 64256, length 0
11:42:10.300123 IP 192.168.1.10.50120 > 142.250.183.36.443: Flags [P.], seq 1:518, ack 1,
                   win 64256, length 517
11:42:10.312345 IP 142.250.183.36.443 > 192.168.1.10.50120: Flags [.], ack 518, win 65535, length 0
11:42:10.400456 IP 142.250.183.36.443 > 192.168.1.10.50120: Flags [P.], seq 1:1461, ack 518,
                   win 65535, length 1460
11:42:10.412567 IP 192.168.1.10.50120 > 142.250.183.36.443: Flags [.], ack 1461, win 64256, length 0

10 packets captured
10 packets received by filter
0 packets dropped by kernel
```

4. ping



```
[user123@localhost ~]$ ping
ping: usage error: Destination address required
[user123@localhost ~]$ ping google.com
PING google.com (192.178.158.100) 56(84) bytes of data.
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=1 ttl=100 time=41.8 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=2 ttl=100 time=41.4 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=3 ttl=100 time=41.1 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=4 ttl=100 time=41.1 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=5 ttl=100 time=40.9 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=6 ttl=100 time=42.7 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=7 ttl=100 time=41.5 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=8 ttl=100 time=42.3 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=9 ttl=100 time=42.4 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=10 ttl=100 time=40.9 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=11 ttl=100 time=41.1 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=12 ttl=100 time=42.1 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=13 ttl=100 time=41.3 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=14 ttl=100 time=41.0 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=15 ttl=100 time=41.7 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=16 ttl=100 time=41.9 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=17 ttl=100 time=41.2 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=18 ttl=100 time=42.2 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=19 ttl=100 time=42.0 ms
64 bytes from lcdels-in-f100.1e100.net (192.178.158.100): icmp_seq=20 ttl=100 time=41.4 ms
                                          158.100): icmp_seq=21 ttl=100 time=40.6 ms
```

5.



```
[user123@localhost ~]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe03:e254  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:03:e2:54  txqueuelen 1000  (Ethernet)
        RX packets 38  bytes 4682 (4.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 75  bytes 7876 (7.6 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 17  bytes 2062 (2.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 17  bytes 2062 (2.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Result:

The windows and the Linux networking commands was successfully executed & implemented.